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Hot spots policing and crime reduction: an update of an ongoing systematic review and meta-analysis

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Abstract

Objectives This updated systematic review assesses the effects of focused police crime prevention interventions at crime hot spots. The review also examined whether focused police actions at specific locations result in crime displacement or diffusion of crime control benefits.

Methods Systematic review protocols and conventions of the Campbell Collaboration were followed to identify eligible hot spots policing studies, and meta-analytic techniques were used to assess the impact of hot spots policing on crime and investigate the influence of moderating variables.

Results The search strategies identified 65 studies containing 78 tests of hot spots policing interventions. Meta-analyses revealed a small statistically significant mean effect size favoring the effects of hot spots policing in reducing crime outcomes at treatment places relative to control places. Crime displacement and diffusion effects were measured in 40 tests. Meta-analyses favored a small statistically significant diffusion of crime control benefits over displacement.

Conclusion The extant evaluation research provides fairly robust evidence that hot spots policing is an effective crime prevention strategy. Focused police intervention at hot spot locations does not seem to result in the spatial displacement of crime into areas immediately surrounding targeted locations. Rather, crime control benefits seem to diffuse into proximate areas.

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Keywords Police · Crime hot spots · Hot spots policing · Meta-analysis · Systematic review · Policing experiment · Crime displacement · Crime prevention

Introduction

Over the past 30 years, crime scholars and practitioners have pointed to the potential benefits of focusing crime prevention efforts on crime places. A number of studies suggest that crime is not spread evenly across city landscapes. Rather, there is significant clustering of crime in small places, or "hot spots," that generate half of all criminal events (Pierce et al. 1988; Sherman et al. 1989a). Even within the most crime-ridden neighborhoods, crime clusters at a few discrete locations and other areas are relatively crime free (Weisburd et al. 2012a). More recent research has reinforced this idea of crime concentrations (Braga et al. 2017) and led Weisburd (2015) to argue that there is a "law of crime concentration" at places showing not just that crime is concentrated but that it is concentrated at similar levels across cities and across time. A number of researchers have argued that many crime problems can be reduced more efficiently if police officers focused their attention on these persistent high-activity crime places (Sherman and Weisburd 1995; Weisburd 1997; Braga and Weisburd 2010). The appeal of focusing limited resources on a small number of high-activity crime places is straightforward. If police can prevent crime at these hot spots, then they might be able to control citywide crime levels (Weisburd et al. 2017a, b).

The available scientific evidence suggests that focused police interventions, such as directed patrols, proactive arrests, and problem-oriented policing, can produce significant crime prevention gains at crime hot spots (see, e.g., Skogan and Frydl 2004; Weisburd and Majimundar 2018). The practice of hot spots policing is supported by two key theoretical crime prevention mechanisms: deterrence and crime opportunity reduction (Nagin et al. 2015; Braga and Schnell 2018). Increasing police visibility in crime hot spots may generate substantial marginal deterrent effects by heightening potential offenders' perceived risk of apprehension and discouraging them from taking advantage of concentrated crime opportunities in these small places (Durlauf and Nagin 2011). Police can also reduce crime by applying problem-oriented policing and situational crime prevention measures to modify opportunity structures that cause crime problems to recur in hot spots (Braga and Weisburd 2010). Critics of place-based interventions, however, charge that such policing strategies result in displacement—that is, criminals move to places not protected by police intervention (e.g., Reppetto 1976). The available evidence suggests that hot spots policing interventions are more likely to be associated with the diffusion of crime control benefits into surrounding areas rather than crime displacement (Clarke and Weisburd 1994; Weisburd et al. 2006; Bowers et al. 2011).

Hot spots policing has become a very popular way for police departments to prevent crime. Many police departments report having the capability to manage and analyze crime data in sophisticated ways and, through management innovations such as Compstat, hold officers accountable for implementing problem-solving strategies to control hot spots locations (Weisburd et al. 2003). The 2007 Law Enforcement Management and Administrative Statistics (LEMAS) survey reported that nearly all police agencies in large metropolitan centers use computers for hot spots identification (Reaves 2010). The Police Executive Research Forum (2008) surveyed 176 US police departments and found that nearly 9 out of 10 agencies used hot spots policing strategies to deal with violent crime in

their jurisdictions and that problem-solving techniques were often deployed to address violent crime hot spots. In a more recent study of a representative sample of police agencies, the National Police Research Platform reported that 75% of the agencies surveyed used the hot spots policing approach (Mastrofski and Fridell n.d., reported in Weisburd and Majimundar 2018).

The widespread use of hot spots policing to prevent crime warrants ongoing careful reviews of the available empirical evidence on the crime control benefits of the approach. If hot spots policing program are effective in controlling crime, the societal benefits may be considerable. For instance, in an influential article, Durlauf and Nagin (2011) suggested that crime and incarceration in the USA would both be reduced if resources were shifted from imprisonment to policing. Among other focused police interventions, they specifically point to evaluations of hot spots policing deployment strategies as evidence that the police, when properly oriented, can prevent crime. As new program evaluations are completed, however, conclusions on the crime control efficacy of hot spots policing could change in response to the growing scientific evidence base. For instance, recent hot spots policing studies have reported null effects (Gerell 2016), crime increases (Phillips et al. 2016), and modest crime displacement (Blattman et al. 2017). This short report provides updated findings from an ongoing Campbell Collaboration systematic review of the effects of hot spots policing on crime (Braga 2001, 2005, 2007; Braga et al. 2012, 2014).

Methods

The protocols and conventions of the Campbell Collaboration were followed in this systematic review and meta-analysis of the scientific evidence on the effects of hot spots policing on crime.[1] Meta-analysis is an analytic method designed to synthesize empirical relationships across studies, such as the effects of a specific crime prevention intervention on criminal offending behavior (Wilson 2001). Meta-analysis uses specialized statistical methods to analyze the relationships between findings and study features (Lipsey and Wilson 2001; Wilson 2001). The "effect size statistic" is the index used to represent the findings of each study in the overall meta-analysis of study findings and represents the strength and direction (positive or negative) of the relationship observed in a particular study (e.g., the size of the treatment effect found). The "mean effect size" represents the average effect of treatment on the outcome of interest across all eligible studies in a particular area and is estimated by calculating a mean that is weighted by the precision of the effect size for each individual study.

Eligibility criteria of studies in the review

To be eligible for this review, interventions used to control crime hot spots were limited to police-led crime control efforts. Suitable police crime control efforts included traditional tactics such as directed patrol and heightened levels of traffic enforcement (Sherman and Weisburd 1995; Ratcliffe et al. 2011), applications of technology such as actively monitored closed circuit televisions (CCTV, see Piza et al. 2015) and license plate readers (Lum et al. 2011a) to enhance police presence in high-crime places, and alternative strategies such as

[1] Additional details on the methods of this updated systematic review are included in the full report to the Campbell Collaboration (Braga et al. 2019).

problem-oriented policing (Braga and Bond 2008; Groff et al. 2015). The units of analysis considered in this review were crime hot spots or high-activity crime "places." These small units included specific locations such as stores and apartment buildings as well as clusters of addresses, street blocks, street segments and street intersections (Eck 1997; Weisburd et al. 2016). However, this review does include evaluation designs that compare changes at larger areal units, such as police beats or census tracts, if the implemented hot spots policing program was clearly focused at specific places within the larger areal unit. For instance, the well-known Kansas City Gun Project quasi-experiment evaluated the effects of increased gun seizures focused at gun hot spots within an 8 by 10 block police beat on gun crime relative to traditional policing services in comparison police beats (Sherman and Rogan 1995a).

In eligible studies, crime places that received the hot spots policing intervention were compared to places that experienced routine levels of traditional police service (i.e. regular levels of patrol, ad hoc investigations, etc.). The comparison group study had to be either experimental or quasi-experimental (nonrandomized) (Shadish et al. 2002). Eligible studies had to measure the effects of police intervention on officially recorded levels of crime at places such as incident reports, citizen emergency calls for service, and arrest data. The review also examined crime displacement and diffusion of crime control benefits reported by the studies (Clarke and Weisburd 1994; Reppetto 1976).

Search strategies for identifying eligible studies

Several strategies were used to perform an exhaustive search for literature fitting the eligibility criteria. First, a keyword search[2] was performed on 15 abstract databases.[3] Second, the bibliographies of past narrative and empirical reviews of literature that examined the effectiveness of police crime control programs were reviewed (e.g., Braga 2016; Higginson and Mazerolle 2014; Johnson et al. 2014; Lum et al. 2011b; Telep et al. 2014a; Telep and Weisburd 2012; Weisburd and Telep 2014; Weisburd et al. 2015a, b). Third, forward searches for works that cited seminal hot spots policing studies were performed (Braga et al. 1999; Braga and Bond 2008; Sherman et al. 1989b; Sherman and Rogan 1995a; Sherman and Weisburd 1995; Weisburd and Green 1995; Weisburd et al. 2006). Fourth, bibliographies of past completed Campbell systematic reviews of police crime prevention efforts were searched (Bowers et al. 2011; Braga 2001, 2005, 2007; Braga et al. 2012, 2014; Braga and Weisburd 2012; Braga et al. 2015; Koper and Mayo-Wilson 2012; Mazerolle et al. 2013). Fifth, hand searches of leading journals in the field were performed.[4]

[2] The terms were: "hot spots AND police", "crime place AND police", "crime clusters AND police", "crime displacement", "place-oriented interventions", "high crime areas AND police", "high crime locations AND police", "targeted policing", "directed patrol", "crackdowns", and "enforcement swamping."

[3] These databases were: Criminal Justice Abstracts, Sociological Abstracts, NCJRS Abstracts, Educational Resources Information Clearinghouse, Google Scholar, Proquest Dissertation and Theses A&I, Westlaw Next, Government Publications Office – Monthly Catalog, Informit, Web of Science Core Collection, Academic Search Premier, C2-SPECTR (original review only), HeinOnline, Social Sciences Premium Collection, and Rutgers University's Gray Literature Database.

[4] These journals were: *Criminology, Criminology & Public Policy, Justice Quarterly, Journal of Research in Crime and Delinquency, Journal of Criminal Justice, Police Quarterly, Policing, Police Practice and Research, British Journal of Criminology, Journal of Quantitative Criminology, Crime & Delinquency, Journal of Criminal Law and Criminology, and Policing & Society*. Hand searches covered January 1979 to February 2017.

The searches were all completed between January 2017 and February 2017. Thus, the review only covers studies completed in 2017 and earlier. Sixth, after finishing the above searches and reviewing the studies as described later, the list of studies meeting our eligibility criteria was emailed to leading criminology and criminal justice scholars knowledgeable in the area of hot spots policing strategies. These 146 experts were defined as those who authored at least one study eligible for this review, anyone involved with the National Academy of Sciences review of police research, and other leading policing scholars (list available in Appendix D of our full report to the Campbell Collaboration). This helped to identify studies the above searches left out as these experts were able to make referrals to studies that were missed, particularly unpublished studies. Finally, an information specialist was engaged at the outset of our review and at points along the way in order to ensure that appropriate search strategies were used to identify the studies meeting the criteria of this review.[5]

Statistical procedures and conventions

As mentioned earlier, meta-analyses were used to determine the size, direction, and statistical significance of the overall impact of hot spots policing strategies on crime by weighting program effect sizes based on the variance of the effect size and the study sample size (Lipsey and Wilson 2001). We used the standardized mean difference effect size (also known as Cohen's *d*; see Cohen 1988) and employed the Effect Size Calculator, developed by David B. Wilson and available on the Campbell Collaboration's website, to calculate standardized mean difference effect sizes for reported outcomes in each study. We then used Biostat's Comprehensive Meta Analysis Version 2.2 to conduct the meta-analysis of effect sizes.

One problem in conducting meta-analyses in crime and justice is that investigators often do not prioritize outcomes examined. This is common in studies in the social sciences in which authors view good practice as demanding that all relevant outcomes be reported. However, the lack of prioritization of outcomes in a study raises the question of how to derive an overall effect of treatment. As such, we analyzed the studies using three approaches. The first combines all reported outcomes reported into an overall average effect size statistic. The second represents the largest effect reported in the studies and gives an upper bound to our findings. It is important to note that in some of the studies with more than one outcome reported, the largest outcome reflected what authors thought would be the most direct program effect. Finally, we present the smallest effect size for each study. This approach is the most conservative and likely underestimates the effect of hot spots policing on crime. We use it here primarily to provide a lower bound to our findings.

Results

Selection and characteristics of eligible studies

Search strategies in the systematic review process generate a large number of citations and abstracts for potentially relevant studies that must be closely screened to determine whether

[5] Ms. Phyllis Schultze of the Gottfredson Library at the Rutgers University School of Criminal Justice.

the studies meet the eligibility criteria (Farrington and Petrosino 2001). The screening process yields a much smaller pool of eligible studies for inclusion in the review. The search strategies produced 26,038 abstracts and the contents of these abstracts were reviewed for any suggestion of an experimental or quasi-experimental evaluation of hot spots policing interventions. A total of 274 distinct abstracts were selected for closer review, and the full-text reports, journal articles, and books for these abstracts were acquired and carefully assessed to determine whether the studies met the eligibility criteria.

The original Campbell systematic review of the effects of hot spots policing on crime identified nine studies (Braga 2001, 2005, 2007) and first update of the review included 19 studies (Braga et al. 2012, 2014). In this iteration, we identified 65 eligible studies. Figure 1 presents the yearly counts of included hot spots policing evaluations and highlights the strong growth in hot spots policing studies since the completion of the previous review. The 46 newly identified studies represented a 242% increase in eligible studies since the prior review. The 65 eligible studies included

(1) Minneapolis Repeat Call Address Policing (RECAP) Program (Sherman et al. 1989a)
(2) New York Tactical Narcotics Teams (Sviridoff et al. 1992)
(3) St. Louis Problem-Oriented Policing in three Drug Market Locations Study (Hope 1994)
(4) Minneapolis Hot Spots Patrol Program (Sherman and Weisburd 1995)
(5) Jersey City Drug Markets Analysis Program (DMAP) (Weisburd and Green 1995)
(6) Kansas City Gun Project (Sherman and Rogan 1995a)
(7) Kansas City Crack House Police Raids Program (Sherman and Rogan 1995b)
(8) Beenleigh Calls for Service Project (Criminal Justice Commission 1998)
(9) Jersey City Problem-Oriented Policing at Violent Places Project (Braga et al. 1999)
(10) Houston Targeted Beat Program (Caeti 1999)
(11) Oakland Beat Health Program (Mazerolle et al. 2000)
(12) Pittsburgh Police Raids at Nuisance Bars Program (Cohen et al. 2003)
(13) Buenos Aires Police Presence after Terror Attack Study (DiTella and Schargrodsky 2004)
(14) Philadelphia Drug Corners Crackdowns Program (Lawton et al. 2005)
(15) Jersey City Displacement and Diffusion Study (Weisburd et al. 2016)
(16) Lowell Policing Crime and Disorder Hot Spots Project (Braga and Bond 2008)
(17) Jacksonville Policing Violent Crime Hot Spots Project (Taylor et al. 2011)
(18) Philadelphia Foot Patrol Experiment (Ratcliffe et al. 2011)
(19) Boston Safe Streets Teams Program (Braga et al. 2011)
(20) DDACTS Program in Washoe County (Beck 2010)
(21) Safer Cities Initiative in Los Angeles (Berk and MacDonald 2010)
(22) License Plate Reader Patrols in Crime Hot Spots in two Adjacent Jurisdictions (Lum et al. 2011a)
(23) Camden 28-Day Crime Suppression Initiative (Ratcliffe and Breen 2011)
(24) Predictive Risk Mapping and Policing in Trafford, Greater Manchester (Fielding and Jones 2012)
(25) Broken Windows Style Crackdowns in three California Cities (Weisburd et al. 2012b)
(26) Operation LASER in Los Angeles (Uchida and Swatt 2013)
(27) Palos Verdes Team Policing Project (Martinez 2013)
(28) License Plate Readers at Crime Hot Spots Experiment in Mesa, Arizona (Koper et al. 2013)

(29) Lowell Smart Policing Initiative (Bond et al. 2014)
(30) DDACTS Program in Shawnee, Kansas (Bryant et al. 2014)
(31) Summer Crime Initiative in Washington, DC (Mazeika 2014)
(32) Operation Impact in Newark, New Jersey (Piza and O'Hara 2014)
(33) St. Louis Metropolitan PD's Firearms Violence Hot Spots Policing Experiment (Rosenfeld et al. 2014)
(34) Hot Spots Randomized Field Trial in Sacramento, California (Telep et al. 2014b)
(35) Trinidad & Tobago Police Services Hotspot Experiment (Sherman et al. 2014)
(36) Policing Crime Hot Spots in Stockholm, Sweden (Marklund and Merenius 2014)
(37) Policing Crime Hot Spots in Eskilstuna, Sweden (Marklund and Merenius 2014)
(38) Anti-Drunk Driving Program in Rajasthan, India (Banerjee et al. 2014)
(39) Philadelphia Policing Tactics Experiment (Groff et al. 2015)
(40) Colorado Springs PD's Risk-Based Intervention (Kennedy et al. 2015)
(41) Newark PD's Risk-Based Intervention (Kennedy et al. 2015)
(42) Kansas City PD's Risk-Based Intervention (Kennedy et al. 2015)
(43) Glendale PD's Risk-Based Intervention (Kennedy et al. 2015)
(44) St. Louis County Hot Spots in Residential Areas Study (Kochel, Burruss, and Weisburd 2015)
(45) Mobile Computing Technology at Crime Hot Spots in a Suburban County (Koper et al. 2015)
(46) Proactive CCTV Monitoring with Directed Police Patrol in Newark, New Jersey (Piza et al. 2015)
(47) Tactical Police Response at Micro-Time Hot Spots (Santos and Santos 2015a, b)
(48) Philadelphia GunStat Model (Sorg 2015)
(49) Dallas Patrol Management Experiment (Weisburd et al. 2015a)
(50) West Midlands Police's Randomized Control Trial of Policing Hot Spots (Williams 2015)
(51) Actively Monitored CCTVs in Stockholm, Sweden (Marklund and Holmberg 2015)
(52) Operation Style in Peterborough, England (Ariel et al. 2016)
(53) Glendale Smart Policing Initiative (Dario 2016)
(54) Policing Violent Crime Hot Spots in Malmö, Sweden (Gerell 2016)
(55) Operation Impact in New York City (MacDonald et al. 2016)
(56) Kansas City Foot Patrol Project (Novak et al. 2016)
(57) Police Paramilitary Raids in Buffalo, New York (Phillips et al. 2016)
(58) Offender-Focused Police Intervention at Hot Spots (Santos and Santos 2016)
(59) New Haven Smart Policing Initiative (Sedelmaier and Hipple 2016)
(60) Operation Menas in London, England (Ariel and Partridge 2016)
(61) Investigating Hot Spots Policing in Copenhagen, Denmark (Attermann 2017)
(62) Hot Spots Policing in Bogotá, Colombia (Blattman et al. 2017)
(63) Philadelphia Predictive Policing Experiment (Ratcliffe et al. 2017)
(64) Flint DDACTS Program (Rydberg et al. 2017)
(65) Operation Strikeforce in Buffalo, New York (Wheeler and Phillips 2017)

Table 1 presents the basic characteristics of the 65 eligible hot spots policing studies. Fifty-one of the 65 (78.5%) identified studies were conducted in the USA. Four hot

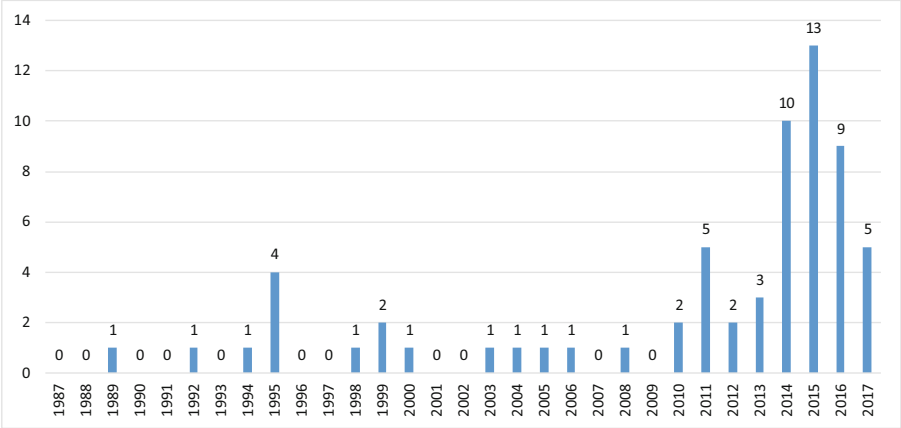

Fig. 1 Number of eligible hot spots policing studies by year ($N = 65$)

Table 1 Characteristics of eligible hot spots policing evaluations

Characteristic	*N*	Percent
Evaluation country (N = 65)		
United States	51	78.5
United Kingdom	4	6.2
Sweden	4	6.2
Other[a]	6	9.2
City population (*N* = 65)		
Small (< 200,000 residents)	12	18.5
Medium (200,000–500,000 residents)	27	41.5
Large (> 500,000 residents)	25	38.5
Small and large	1	1.5
Publication type (*N* = 65)		
Peer-reviewed article	36	55.4
Published report	16	24.6
Thesis/dissertation	7	10.8
Unpublished report/working paper	6	9.2
Evaluation type (*N* = 65)		
Randomized controlled trial	27	41.5
Quasi-experimental	38	58.5
Intervention type (*N* = 78)		
Problem-oriented policing	27	34.6
Increased policing	51	65.4
Displacement/diffusion effects (*N* = 78)		
Measured displacement/diffusion	46	59.0
Did not measure displacement/diffusion	32	41.0

[a] Argentina, Australia, Colombia, Denmark, India, and Trinidad and Tobago

spots policing evaluations were conducted in the United Kingdom and four eligible studies were completed in Sweden. One hot spots policing evaluation was conducted in each of the following countries: Argentina, Australia, Colombia, Denmark, India, and Trinidad & Tobago. Twenty-seven studies (41.5%) were completed in medium-sized cities with between 200,000 and 500,000 residents, 25 studies (38.5%) were completed in large cities with more than 500,000 residents, and 12 studies were completed in smaller cities with less than 200,000 residents (18.5%). One study included both a large and small city in the designated study area (1.5%): Lum et al. (2011a) evaluated the impact of license plate reader technology in crime hot spots located in Alexandria City and the eastern portion of Fairfax County (VA).

Thirty-six of the eligible hot spots policing studies were published in peer-reviewed journals (55.4%), 16 were available as published reports (24.6%), 7 were available as unpublished theses/dissertations (10.8%), and 6 were available as unpublished reports or working papers (9.2%). Twenty-seven eligible studies used randomized controlled trials (41.5%) and 38 eligible studies used quasi-experimental research designs (58.5%) to evaluate the effects of hot spots policing on crime. Eleven of the 65 eligible studies evaluated more than one hot spots policing intervention and carried out these analyses as independent tests. Nine studies examined two separate hot spots policing interventions and two studies examined three hot spots policing interventions. For instance, the seminal Minneapolis RECAP experiment separately evaluated problem-oriented policing interventions at residential and commercial addresses (Sherman et al. 1989b). More recently, Blattman et al. (2017) evaluated the impacts of increased police patrol and, separately, increased police patrol plus municipal services on high-crime street segments in Bogotá, Colombia. In total, the 65 studies included in this review yielded 78 independent experimental and quasi-experimental tests of hot spots policing on crime.

Across the 78 tests of hot spots policing, the specific types of hot spots policing interventions fit broadly into two categories: problem-oriented policing and increased traditional policing. More than one third of hot spots policing programs focused primarily on reducing crime opportunities at places by engaging strategies consistent with problem-oriented policing ($N = 27$, 34.6%). In these studies, the problem-oriented policing strategies generally attempted to change the underlying conditions and situational dynamics that caused problems to recur in high-activity crime places (Goldstein 1990; Braga 2008). Increased traditional policing was used in two-thirds of the eligible hot spots policing ($N = 51$, 65.4%). These programs were generally designed to deter offenders from committing crimes in hot spots areas by increasing police presence and enforcement activities. This was most commonly attempted through increased foot or vehicle patrol ($N = 31$), drug enforcement operations ($N = 6$), offender-focused apprehension programs ($N = 4$), actively monitored CCTV with directed patrol ($N = 3$), and other kinds of increased enforcement activities ($N = 7$).[6] Crime displacement and diffusion of crime control benefits effects were assessed for 46 of the 78 tests of hot spots policing (58.9%).

[6] These activities included roadblocks, patrol with license plate reader technology, zero-tolerance policing, and increased gun searches and seizures.

A noteworthy majority of the hot spots policing evaluations concluded that hot spots policing programs generated significant crime control benefits in the treatment areas relative to the control areas.[7] Only 16 of the 78 (20.5%) tests of hot spots policing interventions did not report noteworthy crime control gains associated with the approach. Sixty-two of the 78 tests (79.5%) of hot spots policing interventions in the 65 eligible studies reported noteworthy crime control gains.

Meta-analysis of the effects of hot spots policing on crime

Our meta-analyses of the effects of hot spots policing programs on crime were limited to 62 of the 65 eligible studies. Two studies, the St. Louis Problem-Oriented Policing in Three Drug Market Locations Study (Hope 1994) and the Beenleigh (Australia) Calls for Service Project (Criminal Justice Commission 1998), did not report the necessary information to calculate program effect sizes. The Houston (TX) Targeted Beat Program (Caeti 1999) did not use appropriate statistical methods to estimate program effects and, unfortunately, accurate effect sizes could not be calculated. We were able to calculate effect sizes for 73 main effects tests and 40 displacement and diffusion tests in these 65 eligible studies.

Using the overall mean effect size from each study for 73 main effects tests, the forest plot in Fig. 2 shows the standardized difference in means between the treatment and control or comparison conditions (effect size) with a 95% confidence interval plotted around them for all tests. Points plotted to the right of 0 indicate a treatment effect; in this case, the test showed a reduction in crime or disorder. Points to the left of 0 indicate a backfire effect where control conditions improved relative to treatment conditions. A random effects model was used to estimate the overall mean effect size based on an a priori assumption of a heterogeneous distribution of effect sizes.[8] The meta-analysis of effect sizes suggests an effect in favor of hot spots policing strategies. Notably, the overall effect size for these studies is .132 ($p < .001$); this would be considered a small mean effect size (see Cohen 1988).

Fifty-seven tests reported effect sizes that favored treatment conditions over control conditions (78.1% of 73 total tests).[9] For the largest effect size meta-analysis, the overall standardized mean difference effect size was .197 and statistically significant at the $p < .05$ level. For the smallest effect size meta-analysis, the overall standardized mean difference effect size was .104 and statistically significant at the $p < .05$ level. Table 2 presents mean effect sizes for the effects of hot spots policing programs on violent crime, property crime, drug offense, and disorder offense outcomes. Hot spots policing programs produced statistically significant ($p < .05$) positive mean effect sizes

[7] Readers interested in reviewing the details of the eligible studies should consult the full report to the Campbell Collaboration. In that report, Table 2 summarizes the treatments, hot spots definitions, and research designs. Table 3 summarizes the main effects of the intervention on crime and disorder measures, treatment effects as measured by other non-official data sources, and, if measured, the immediate spatial displacement and diffusion of crime control benefits effects. A narrative review of the 65 hot spots policing studies and the 78 tests contained in the eligible studies is provided in Appendix C. (see Braga et al. 2019).

[8] For the overall main effects meta-analysis, Q = 362.714, df = 72, $p < 0.001$ and $I^2 = 80.150$.

[9] Random effects models were used to estimate the overall standardized mean effect sizes. For the largest effect size meta-analysis, Q = 437.268, df = 72, $p < 0.001$, $I^2 = 83.534$. For the smallest effect size meta-analysis, Q = 431.914, df = 72, $p < 0.001$, $I^2 = 83.330$.

Study name	Outcome	Std diff in means	Standard error
Trafford Burglary	Burglary incidents	0.977	0.207
KC Gun	Gun crimes	0.866	0.275
Phila. Drug Corners	Combined	0.855	0.258
Port St. Lucie Tactical	Combined	0.718	0.178
St. Louis Patrol	Calls for service	0.630	0.244
LA LASER	Total crime incidents	0.618	0.232
Buenos Aires Police	Motor vehicle theft incidents	0.617	0.169
St. Louis Gun E-Patrol	Combined	0.532	0.255
JC Disp. Prost.	Prostitution events	0.525	0.149
St. Louis POP	Calls for service	0.476	0.244
DDACTS Flint	Violent crime incidents	0.459	0.542
JC Disp. Drug	Drug events	0.441	0.131
Minn. RECAP Resid.	Total calls	0.369	0.132
SCI Los Angeles	Combined	0.355	0.114
Newark Foot Patrol	Violent incidents	0.352	0.148
Sacramento Patrol	Combined	0.345	0.311
Boston SST	Total violent incidents	0.341	0.020
Eskilstuna POP	Assault incidents	0.326	0.197
Rajasthan Checkpoints	Combined	0.316	0.149
Stockholm POP	Robbery incidents	0.308	0.197
Oakland Beat Health	Drug calls	0.279	0.056
Camden Suppression	Total crime incidents	0.276	0.161
Newark A-CCTV	Combined	0.270	0.223
Port St. Lucie Offender	Crime incidents	0.251	0.290
Mesa LPR Patrol	Combined	0.242	0.245
DDACTS Kansas	Total crime incidents	0.241	0.094
Bogota Patrol / Municipal	Crime incidents	0.239	0.117
Newark RTM	Gun violence incidents	0.237	0.174
New Haven SPI	Total crime incidents	0.236	0.196
St. Louis Gun Patrol	Combined	0.234	0.251
Phila. Tactics Offender	Violent crime incidents	0.225	0.071
Colorado Springs RTM	Auto theft incidents	0.223	0.156
Peterborough Patrol	Combined	0.201	0.237
Suburban MCT	Total crime incidents	0.189	0.152
Dallas AVL Patrol	Total crime incidents	0.176	0.063
JC DMAP	Combined	0.147	0.270
Lowell POP	Total calls	0.145	0.034
JC POP	Combined	0.143	0.043
Phila. Foot Patrol	Violent incidents	0.143	0.021
NY Impact	Crime incidents	0.132	0.009
Mesa Patrol / Surv.	Combined	0.130	0.244
Pittsburgh Bar Raids	Drug calls	0.125	0.038
West Midlands Patrol	Combined	0.122	0.190
T&T Directed Patrol	Violent crime incidents	0.106	0.032
Buffalo Strikeforce	Combined	0.099	0.078
NYC TNT 67	Combined	0.087	0.077
Glendale SPI	Calls for service	0.082	0.019
Kansas City RTM	Agg. violence incidents	0.072	0.156
Lowell SPI	Property incidents	0.072	0.050
Minn. Patrol	Total calls	0.061	0.015
KC Crack	Total calls	0.051	0.039
Buffalo Raids	Total calls	0.050	0.117
KC Foot Patrol	Violent crime incidents	0.036	0.055
Bogota Patrol	Crime incidents	0.031	0.046
Copenhagen POP	Total crime incidents	0.028	0.033
Phila. Tactics Foot	Violent crime incidents	0.020	0.059
Minn. RECAP Comm.	Total calls	0.015	0.137
N. VA LPR	Total crime incidents	0.004	0.365
Jacksonville POP	Combined	-0.005	0.092
London Bus Stops	Combined	-0.006	0.200
Phila. Pred. Marked	Combined	-0.015	0.067
DC Summer Crime	Robbery calls	-0.024	0.138
NYC TNT 70	Combined	-0.027	0.080
Suburban LA disorder	Combined	-0.027	0.193
Phila. Tactics POP	Violent crime incidents	-0.048	0.059
Jacksonville Patrol	Combined	-0.055	0.096
Phila. Pred. Unmarked	Combined	-0.056	0.065
Sweden A-CCTV	Violent crime incidents	-0.062	0.043
DDACTS Nevada	Combined	-0.067	0.117
Malmo A-CCTV	Street assaults	-0.134	0.235
Glendale RTM	Agg. violence incidents	-0.181	0.221
Phila. GunStat	Violent crime incidents	-0.199	0.079
Palos Verde Team	Total calls	-0.321	0.106
		0.132	0.017

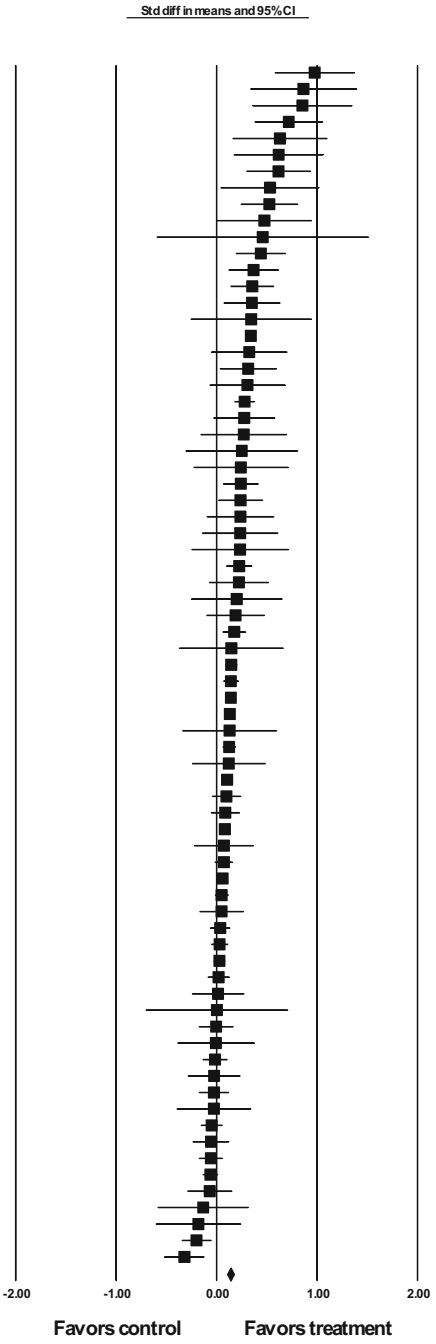

Fig. 2 Combined effect sizes for study outcomes

Table 2 The effects of hot spots policing on specific crime types

Crime category	*N* studies	Effect size	95% C.I.
Violent crimes	44	.102*	.047, .156
Property crimes	26	.124*	.063, .185
Disorder offenses	15	.161*	.101, .220
Drug offenses	10	.244*	.114, .373

Note: Random effects meta-analysis models used in all reported effect sizes
*$p < .05$

for violent crime outcomes (.102), property crime outcomes (.124), disorder outcomes (.161), and drug crime outcomes (.244).

Meta-analysis of crime displacement and diffusion effects

Crime displacement and diffusion effects were analyzed jointly as two sides of a single distribution that ranged from harmful to beneficial effects. Using the overall mean effect size from each study for 40 displacement and diffusion tests, the forest plots in Fig. 3 show the standardized difference in means between the treatment and control or comparison conditions (effect size) with a 95% confidence interval plotted around them for all tests. Points plotted to the right of 0 indicate a diffusion of crime control benefits effect; in this case, the test showed a reduction in crime or disorder in the areas surrounding the targeted hot spots. Points to the left of 0 indicate a crime displacement effect. We used a random effects model to estimate the overall mean effect size.[10] The meta-analysis suggests a small but statistically significant overall diffusion of crime control benefits effect (.086) generated by the hot spots policing strategies ($p < .001$).

Evaluation design and program type as effect size moderators

Given the important distinction in methodological quality between the randomized controlled trials and quasi-experimental evaluation studies, we also examined research design as a moderator variable. It is well known among social scientists that program evaluations with more rigorous research designs tend to report null effects (see, e.g., Rossi 1987). Figure 4 presents a random effects model examining the two different classes of evaluation designs included in this review.[11] Prior research has suggested that weaker designs are more likely to report stronger effects in crime and justice studies (Weisburd et al. 2001; Welsh et al. 2011). However, the purported relationship between quasi-experimental designs and larger effect sizes has not been universally found. Other studies have found that well-designed quasi-experiments can produce effect size similar

[10] Random effects models were used to estimate the overall displacement and diffusion standardized mean effect sizes: Q = 22850.673, df = 39, $p < 0.001$, $I^2 = 99.829$.

[11] We used a random effects model for this comparison. For the quasi-experiments, Q = 267.626, df = 37, $p < 0.001$, $I^2 = 86.175$. For the randomized controlled trials, Q = 69.379, df = 34, $p < 0.001$, $I^2 = 50.994$. For the overall analysis, the Between Group Q = 8.159, df = 1, $p < 0.004$, suggesting that the type of evaluation produced statistically-significant differences in observed crime outcomes. The moderated overall effect size was .128 (Standard error = 0.017, $p < .001$, 95% CI = .094, .162).

Study name	Outcome	Std diff in means	Standard error
Phila. Drug Corners	Combined	0.580	0.065
JC Disp. Prost.	Combined	0.395	0.019
SCI Los Angeles	Crime incidents	0.390	0.114
Port St. Lucie Tactical	Combined	0.363	0.208
Newark Foot Patrol	Violent crime incidents	0.352	0.148
Malmo A-CCTV	Street assault incidents	0.272	0.256
Bogota Patrol / Municipal	Crime incidents	0.239	0.118
Phila. Tactics Offender	Violent crime incidents	0.191	0.083
KC Foot Patrol	Violent incidents	0.189	0.055
London Bus Stops	Combined	0.167	0.200
JC DMAP	Combined	0.161	0.269
Glendale RTM	Agg. violence incidents	0.161	0.150
Oakland Beat Health	Drug calls	0.160	0.035
JC Disp. Drug	Combined	0.124	0.015
Colorado Springs RTM	MV theft incidents	0.120	0.118
Newark RTM	Gun violence incidents	0.120	0.106
Stockholm POP	Robbery incidents	0.109	0.196
DC Summer Crime	Robbery calls	0.104	0.183
Camden Suppression	Crime incidents	0.102	0.120
Phila. Pred. Marked	Property incidents	0.088	0.066
NY Impact	Crime incidents	0.074	0.008
Buenos Aires Police	Combined	0.051	0.082
JC POP	Combined	0.049	0.001
St. Louis Gun E-Patrol	Gun violence incidents	0.025	0.250
Newark A-CCTV	Combined	0.019	0.199
Lowell POP	Total calls	0.013	0.001
Peterborough Patrol	Crime incidents	0.010	0.046
Boston SST	Violent incidents	0.009	0.000
Glendale SPI	Calls for service	0.006	0.063
Phila. Pred. Unmarked	Property incidents	-0.015	0.066
Bogota Patrol	Crime incidents	-0.025	0.047
KC Gun	Gun crimes	-0.044	0.263
Jacksonville POP	Combined	-0.050	0.167
Phila. Foot Patrol	Violent incidents	-0.057	0.000
Sweden A-CCTV	Violent incidents	-0.075	0.057
Mesa LPR Patrol	Combined	-0.077	0.245
Jacksonville Patrol	Combined	-0.088	0.196
Sacramento Patrol	Combined	-0.157	0.100
Palos Verde Team	Calls for service	-0.163	0.102
Mesa Patrol / Surv.	Combined	-0.248	0.244
		0.086	0.012

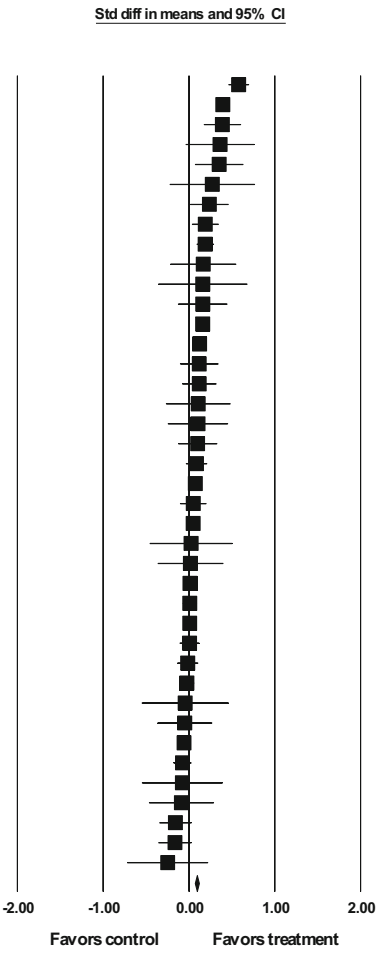

Fig. 3 Combined effects sizes for displacement and diffusion outcomes

in magnitude to randomized experiments (e.g., see Lipsey and Wilson 2001; Shadish and Ragsdale 1996). In this review, the quasi-experimental designs were associated with a larger within-group effect size (.171, $p < .001$) relative to the randomized controlled trial designs (.109, $p < .001$).

Our narrative review documented two general types of programs that represented fundamentally different orientations in dealing with the problems of high-activity crime places. Problem-oriented policing programs attempt to change the underlying conditions at hot spots that cause them to generate recurring crime problems (Goldstein 1990; Braga and Weisburd 2010). The other hot spots policing interventions applied increased traditional policing activities concentrated at specific places to prevent crime through general deterrence and increased risk of apprehension (Nagin et al. 2015).

Group by Design	Study name	Outcome	Std diff in means	Standard error
Quasi	Trafford Burglary	Burglary incidents	0.977	0.207
Quasi	KC Gun	Gun crimes	0.866	0.275
Quasi	Phila. Drug Corners	Combined	0.855	0.258
Quasi	Port St Lucie Tactical	Combined	0.718	0.178
Quasi	LA LASER	Total crime incidents	0.618	0.232
Quasi	Buenos Aires Police	Motor vehicle theft incidents	0.617	0.169
Quasi	JC Disp. Prost.	Prostitution events	0.525	0.149
Quasi	DDACTS Flint	Violent crime incidents	0.459	0.542
Quasi	New Haven SPI	Combined	0.451	0.256
Quasi	JC Disp. Drug	Drug events	0.441	0.131
Quasi	SCI Los Angeles	Combined	0.355	0.114
Quasi	Newark Foot Patrol	Violent incidents	0.352	0.148
Quasi	Boston SST	Total violent incidents	0.341	0.020
Quasi	Eskilstuna POP	Assault incidents	0.326	0.197
Quasi	Stockholm POP	Robbery incidents	0.308	0.197
Quasi	Camden Suppression	Combined	0.291	0.272
Quasi	DDACTS Kansas	Combined	0.269	0.202
Quasi	Port St Lucie Offender	Crime incidents	0.251	0.290
Quasi	Newark RTM	Gun violence incidents	0.237	0.174
Quasi	Colorado Springs RTM	Auto theft incidents	0.223	0.156
Quasi	Pittsburgh Bar Raids	Drug calls	0.125	0.038
Quasi	West Midlands Patrol	Combined	0.122	0.190
Quasi	NY Impact	Combined	0.114	0.012
Quasi	Buffalo Strikeforce	Combined	0.099	0.078
Quasi	NYC TNT 67	Combined	0.087	0.077
Quasi	Glendale SPI	Calls for service	0.082	0.019
Quasi	Kansas City RTM	Agg. violence incidents	0.072	0.156
Quasi	Lowell SPI	Property incidents	0.072	0.050
Quasi	KC Foot Patrol	Violent crime incidents	0.036	0.055
Quasi	Buffalo Raids	Combined	0.035	0.117
Quasi	DC Summer Crime	Robbery calls	-0.024	0.138
Quasi	NYC TNT 70	Combined	-0.027	0.080
Quasi	Sweden A-CCTV	Violent crime incidents	-0.062	0.043
Quasi	DDACTS Nevada	Combined	-0.067	0.117
Quasi	Malmo A-CCTV	Street assaults	-0.134	0.235
Quasi	Glendale RTM	Agg. violence incidents	-0.181	0.221
Quasi	Phila. GunStat	Violent crime incidents	-0.199	0.079
Quasi	Palos Verde Team	Total calls	-0.321	0.106
Quasi			0.171	0.031
RCT	St. Louis Patrol	Calls for service	0.630	0.244
RCT	St. Louis Gun E-Patrol	Combined	0.532	0.255
RCT	St. Louis POP	Calls for service	0.476	0.244
RCT	Minn. RECAP Resid.	Total calls	0.369	0.132
RCT	Sacramento Patrol	Combined	0.345	0.311
RCT	Rajasthan Checkpoints	Combined	0.316	0.149
RCT	Oakland Beat Health	Drug calls	0.279	0.056
RCT	Newark A-CCTV	Combined	0.270	0.223
RCT	Mesa LPR Patrol	Combined	0.242	0.245
RCT	Bogota Patrol / Municipal	Crime incidents	0.239	0.117
RCT	St. Louis Gun Patrol	Combined	0.234	0.251
RCT	Phila. Tactics Offender	Violent crime incidents	0.225	0.071
RCT	JC POP	Combined	0.217	0.122
RCT	Peterborough Patrol	Combined	0.201	0.237
RCT	Lowell POP	Combined	0.195	0.125
RCT	Suburban MCT	Total crime incidents	0.189	0.152
RCT	Dallas AVL Patrol	Total crime incidents	0.176	0.063
RCT	JC DMAP	Combined	0.147	0.270
RCT	Phila. Foot Patrol	Violent incidents	0.143	0.021
RCT	Mesa Patrol / Surv.	Combined	0.130	0.244
RCT	T&T Directed Patrol	Violent crime incidents	0.106	0.032
RCT	Copenhagen POP	Combined	0.083	0.109
RCT	KC Crack	Combined	0.072	0.097
RCT	Minn. Patrol	Combined	0.054	0.021
RCT	Bogota Patrol	Crime incidents	0.031	0.046
RCT	Phila. Tactics Foot	Violent crime incidents	0.020	0.059
RCT	Minn. RECAP Comm.	Total calls	0.015	0.137
RCT	London Bus Stops	Combined	-0.006	0.200
RCT	Jacksonville POP	Combined	-0.012	0.102
RCT	Phila. Pred. Marked	Combined	-0.015	0.067
RCT	Suburban LA disorder	Combined	-0.027	0.193
RCT	Phila. Tactics POP	Violent crime incidents	-0.048	0.059
RCT	Phila. Pred. Unmarked	Combined	-0.056	0.065
RCT	Jacksonville Patrol	Combined	-0.067	0.108
RCT	N. VA LPR	Combined	-0.082	0.365
RCT			0.109	0.021
Overall			0.128	0.017

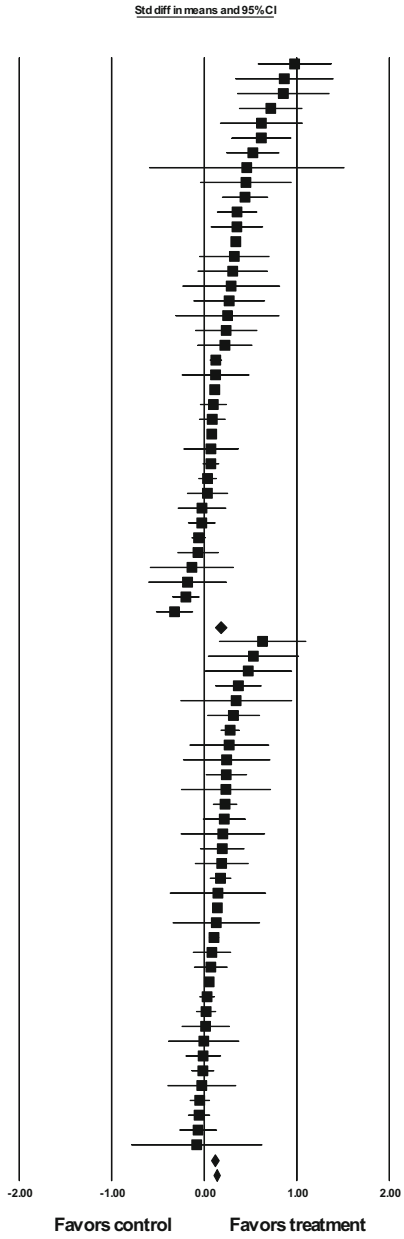

Fig. 4 Research Design Type as Moderator for Study Outcomes

There is, of course, some overlap between the enforcement interventions employed by the problem-oriented policing hot spots programs and the actions taken by the increased policing hot spots programs. Nonetheless, program type represented a potentially

influential moderator of the observed effect sizes in our overall meta-analysis. Figure 5 presents a random effects model examining the two different hot spots policing program types: problem-oriented policing and increased policing.[12] Our meta-analysis revealed that problem-oriented policing programs produced a modestly larger overall mean effect size (.164, $p < .001$) relative to the size of the overall mean effect size generated by increased traditional policing programs (.108, $p < .001$).

Publication bias

Publication bias, generally defined as the concern that the collection of studies easily available to a reviewer represents those studies most likely to have statistically significant results, presents a strong challenge to any review of evaluation studies (Rothstein 2008). The credibility of a review arguably depends more heavily on the collection of studies reviewed than on which statistical methods of synthesis are used (Wilson 2009). Similar to the problem of a biased study sample leading to biased results in an individual study, a biased collection of studies will potentially lead to biased conclusions in a systematic review (Rothstein and Hopewell 2009). Our search strategies were explicitly designed to mitigate the potential effects of publication bias on our analyses. Indeed, it is encouraging that nearly half of the eligible studies (29 of 65, 44.6%) were acquired through gray literature sources such as published reports, theses, dissertations, unpublished reports, and unpublished working papers. The studies identified through gray literature sources reported a much smaller overall mean effect size (.060, $p < .001$) when compared to the overall mean effect size (.200, $p < .001$) reported by studies in published journal articles, suggesting that our search strategies were successful in identifying a range of hot spots policing studies with varying effects on crime outcomes.[13]

Like many systematic reviews, our meta-analyses used the trim-and-fill procedure to explore whether publication bias might be affecting the results and to estimate how the reported effects would change if the bias were to be removed (Duval 2005; Duval and Tweedie 2000). The diagnostic funnel plot is based on the idea that, in the absence of bias, the plot of study effect sizes should be symmetric about the mean effect size. If there is asymmetry, the trim-and-fill procedure imputes the missing studies, adds them to the analysis, and then re-computes the mean effect size. Trim-and-fill procedures do suffer from some well-known limitations that could result in the underestimation or overestimation of publication bias (Rothstein 2008; Simonsohn et al. 2014). Nonetheless, this approach does provide reviewers with a well-understood measure of the possible influence of bias on their meta-analytic results.

[12] For problem-oriented policing programs, Q = 179.543, df = 24, $p < 0.001$, $I^2 = 86.632$. For increased policing programs, Q = 162.328, df = 47, $p < 0.001$, $I^2 = 71.046$. The between Q = 20.852, df = 1, $p < 0.001$, suggesting that the hot spots policing program type produced statistically-significant differences in observed crime outcomes. The moderated overall effect size was .120 (Standard error = 0.017, $p < .001$, 95% CI = .086, .153).

[13] The 29 gray literature studies included 32 independent tests of hot spots policing programs and the 36 journal article studies included 41 independent tests of hot spots policing programs. For gray literature studies, Q = 73.908, df = 31, $p < 0.001$, $I^2 = 58.056$. For journal article studies, Q = 228.913, df = 40, $p < 0.001$, $I^2 = 82.526$. The between Q = 42.342, df = 1, $p < 0.001$, suggesting that the publication type produced statistically-significant differences in observed crime outcomes. The moderated overall effect size was .125 (Standard error = 0.018, $p < .001$, 95% CI = .089, .161).

Group by Program Type	Study name	Outcome	Std diff in means	Standard error
Increased Policing	Trafford Burglary	Burglary incidents	0.977	0.207
Increased Policing	KC Gun	Gun crimes	0.866	0.275
Increased Policing	Phila. Drug Corners	Combined	0.855	0.258
Increased Policing	Port St. Lucie Tactical	Combined	0.718	0.178
Increased Policing	St. Louis Patrol	Calls for service	0.630	0.244
Increased Policing	LA LASER	Total crime incidents	0.618	0.232
Increased Policing	Buenos Aires Police	Motor vehicle theft incidents	0.617	0.169
Increased Policing	St. Louis Gun E-Patrol	Combined	0.532	0.255
Increased Policing	DDACTS Flint	Violent crime incidents	0.459	0.542
Increased Policing	Newark Foot Patrol	Violent incidents	0.352	0.148
Increased Policing	Sacramento Patrol	Combined	0.345	0.311
Increased Policing	Rajasthan Checkpoints	Combined	0.316	0.149
Increased Policing	Camden Suppression	Total crime incidents	0.276	0.161
Increased Policing	Newark A-CCTV	Combined	0.270	0.223
Increased Policing	Port St. Lucie Offender	Crime incidents	0.251	0.290
Increased Policing	Mesa LPR Patrol	Combined	0.242	0.245
Increased Policing	DDACTS Kansas	Total crime incidents	0.241	0.094
Increased Policing	St. Louis Gun Patrol	Combined	0.234	0.251
Increased Policing	Phila. Tactics Offender	Violent crime incidents	0.225	0.071
Increased Policing	Peterborough Patrol	Combined	0.201	0.237
Increased Policing	Suburban MCT	Total crime incidents	0.189	0.152
Increased Policing	Dallas AVL Patrol	Total crime incidents	0.176	0.063
Increased Policing	Phila. Foot Patrol	Violent incidents	0.143	0.021
Increased Policing	NY Impact	Crime incidents	0.132	0.009
Increased Policing	Mesa Patrol / Surv.	Combined	0.130	0.244
Increased Policing	Pittsburgh Bar Raids	Drug calls	0.125	0.038
Increased Policing	West Midlands Patrol	Combined	0.122	0.190
Increased Policing	T&T Directed Patrol	Violent crime incidents	0.106	0.032
Increased Policing	Buffalo Strikeforce	Combined	0.099	0.078
Increased Policing	NYC TNT 67	Combined	0.087	0.077
Increased Policing	Minn. Patrol	Total calls	0.061	0.015
Increased Policing	KC Crack	Total calls	0.051	0.039
Increased Policing	Buffalo Raids	Total calls	0.050	0.117
Increased Policing	KC Foot Patrol	Violent crime incidents	0.036	0.055
Increased Policing	Bogota Patrol	Crime incidents	0.031	0.046
Increased Policing	Phila. Tactics Foot	Violent crime incidents	0.020	0.059
Increased Policing	N. VA LPR	Total crime incidents	0.004	0.365
Increased Policing	London Bus Stops	Combined	-0.006	0.200
Increased Policing	Phila. Pred. Marked	Combined	-0.015	0.067
Increased Policing	DC Summer Crime	Robbery calls	-0.024	0.138
Increased Policing	NYC TNT 70	Combined	-0.027	0.080
Increased Policing	Jacksonville Patrol	Combined	-0.055	0.096
Increased Policing	Phila. Pred. Unmarked	Combined	-0.056	0.065
Increased Policing	Sweden A-CCTV	Violent crime incidents	-0.062	0.043
Increased Policing	DDACTS Nevada	Combined	-0.067	0.117
Increased Policing	Malmo A-CCTV	Street assaults	-0.134	0.235
Increased Policing	Glendale RTM	Agg. violence incidents	-0.181	0.221
Increased Policing	Phila. Gun Stat	Violent crime incidents	-0.199	0.079
Increased Policing			0.108	0.019
Problem-Oriented	JC Disp. Prost.	Prostitution events	0.525	0.149
Problem-Oriented	St. Louis POP	Calls for service	0.476	0.244
Problem-Oriented	JC Disp. Drug	Drug events	0.441	0.131
Problem-Oriented	Minn. RECAP Resid.	Total calls	0.369	0.132
Problem-Oriented	SCI Los Angeles	Combined	0.355	0.114
Problem-Oriented	Boston SST	Total violent incidents	0.341	0.020
Problem-Oriented	Eskilstuna POP	Assault incidents	0.326	0.197
Problem-Oriented	Stockholm POP	Robbery incidents	0.308	0.197
Problem-Oriented	Oakland Beat Health	Drug calls	0.279	0.056
Problem-Oriented	Bogota Patrol / Municipa	Crime incidents	0.239	0.117
Problem-Oriented	Newark RTM	Gun violence incidents	0.237	0.174
Problem-Oriented	New Haven SPI	Total crime incidents	0.236	0.196
Problem-Oriented	Colorado Springs RTM	Auto theft incidents	0.223	0.156
Problem-Oriented	JC DMAP	Combined	0.147	0.270
Problem-Oriented	Lowell POP	Total calls	0.145	0.034
Problem-Oriented	JC POP	Combined	0.143	0.043
Problem-Oriented	Glendale SPI	Calls for service	0.082	0.019
Problem-Oriented	Kansas City RTM	Agg. violence incidents	0.072	0.156
Problem-Oriented	Lowell SPI	Property incidents	0.072	0.050
Problem-Oriented	Copenhagen POP	Total crime incidents	0.028	0.033
Problem-Oriented	Minn. RECAP Comm.	Total calls	0.015	0.137
Problem-Oriented	Jacksonville POP	Combined	-0.005	0.092
Problem-Oriented	Suburban LA disorder	Combined	-0.027	0.193
Problem-Oriented	Phila. Tactics POP	Violent crime incidents	-0.048	0.059
Problem-Oriented	Palos Verde Team	Total calls	-0.321	0.106
Problem-Oriented			0.164	0.037
Overall			0.120	0.017

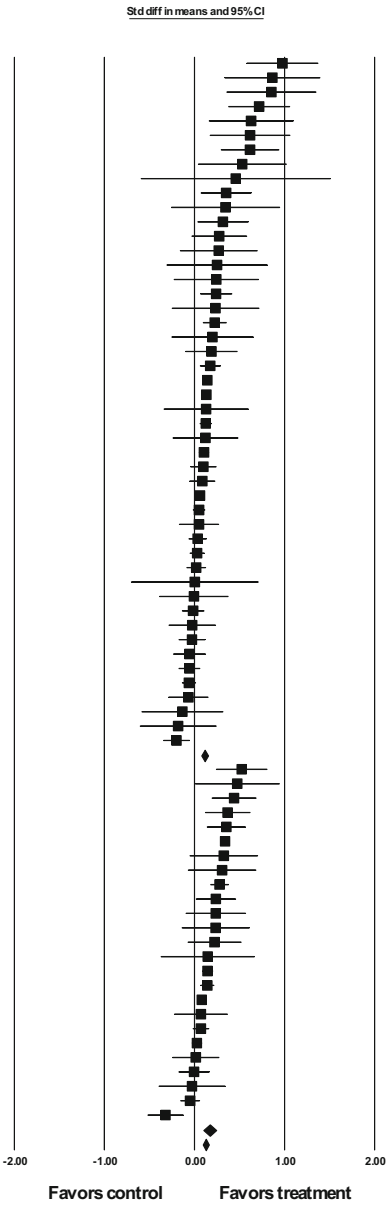

Fig. 5 Program Type as Moderator for Study Outcomes

A visual inspection of the resulting funnel plot indicated some asymmetry with more studies with a large effect and a large standard error to the right of the mean than the left of the mean. The trim-and-fill procedure determined that 11 studies should be added to create symmetry. The funnel plot with imputed studies is presented in Fig. 6. Using a

random effects model, the mean random effect decreased from 0.132 (95% *CI* = 0.097, 0.165) to 0.103 (95% *CI* = 0.067, 0.138). Indeed, the 95% confidence intervals substantially overlap, suggesting that the underlying parameters may not be different. Nevertheless, the trim-and-fill result suggests mild publication selection bias. However, the adjusted mean effect size remained a similar statistically-significant small size and, as such, the observed publication bias does not appear to be sufficient to nullify the results (as suggested by the funnel plot in Fig. 6).

Conclusion

The results of this ongoing systematic review once again support the assertion that focusing police efforts at high activity crime places can be effective in preventing crime (Skogan and Frydl 2004; Weisburd and Majimundar 2018). The updated systematic review identified 78 tests of hot spots policing in 65 eligible studies. Sixty-two of the 78 tests reported noteworthy crime control gains associated with the hot spots policing interventions when treatment conditions were compared to control conditions. A meta-analysis of key reported outcome measures revealed a small but statistically significant mean effect size favoring the effects of hot spots policing in reducing crime in treatment places relative to control places. This small crime reduction effect was observed in both randomized experiments and quasi-experiments and across a diverse range of crime outcomes. When crime displacement was measured, it was very limited and unintended crime prevention benefits were more likely to be associated with the hot spots policing programs (see also Bowers et al. 2011). A meta-analysis of key reported outcome measures in 40 tests revealed a small but statistically significant mean effect size

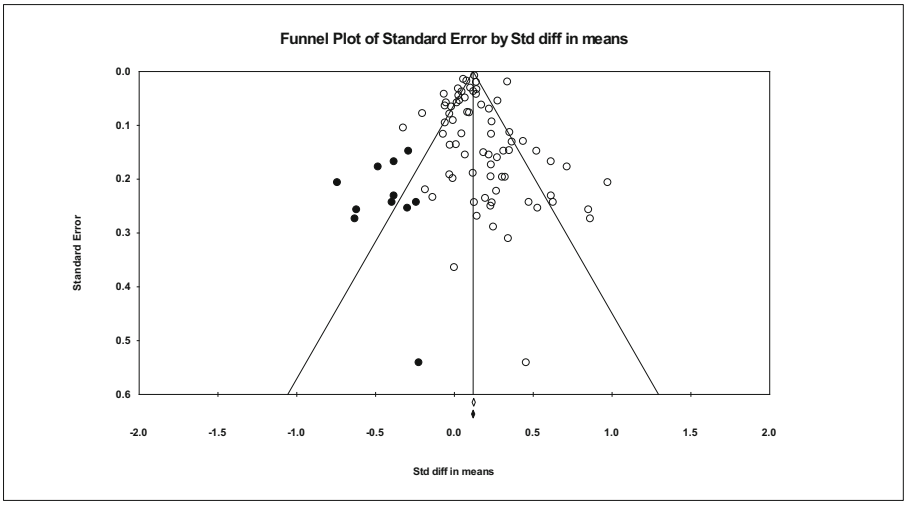

Fig. 6 The funnel plot with imputed studies. Note: Empty circles are the original studies. Filled-in circles indicate 11 imputed studies from the trim-and-fill analysis. These additional studies only slightly changed the mean effect size estimate. Using a random effects model, decreased from 0.132 (95% *CI* = 0.097, 0.165) to 0.103 (95% *CI* = 0.067, 0.138)

favoring a diffusion of crime control benefits rather than a crime displacement effect.The updated meta-analysis found that problem-oriented policing interventions seemed to generate moderately larger overall effect sizes when compared to the increased policing interventions. While increasing presence and concentrating traditional enforcement activities constitute an effective police response to crime hot spots, it seems likely that altering place characteristics and dynamics will produce larger crime prevention benefits (Braga and Weisburd 2010). We believe that the problem-oriented policing approach holds great promise in developing tailored responses to very specific recurring problems at crime hot spots. While it is difficult for police agencies to implement the "ideal" version of problem-oriented policing (Cordner and Biebel 2005; Eck 2006), the available evidence suggests that even "shallow" problem solving better focuses police crime prevention efforts at crime hot spots (Braga and Weisburd 2006).

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